35—BAR SCALES (continued)

BAR SCALE CALCULATIONS — MILES (1 mile = 63,360 inches)

FRACTIONAL SCALE	SCALE TO MAP REPRESENTATION		TO FIND MILES PER INCH (x in ratio)	MILES PER INCH	TOTAL MILES ON SCALE	TO FIND TOTAL SCALE LENGTH IN INCHES (y in ratio)	TOTAL SCALE LENGTH (INCHES)
	Scale Unit :	represents Map Unit	Use ratio below or $\frac{\text{SCALE}}{63\,360}$			Use ratio below or $\frac{\text{Miles on scale}}{\text{Miles per inch}}$	
1:12 000	1 inch	12 000 in	$\frac{63\,360}{1} = \frac{12\,000}{x}$	0.1893939	1.5	$\frac{0.1893939}{1} = \frac{1.5}{y}$	7.920
1:20 000	1 inch	20 000 in	$\frac{63\,360}{1} = \frac{20\,000}{x}$	0.3156565	2	$\frac{0.3156565}{1} = \frac{2}{y}$	6.336
1:24 000	1 inch	24 000 in	$\frac{63\,360}{1} = \frac{24\,000}{x}$	0.3787878	2	$\frac{0.3787878}{1} = \frac{2}{y}$	5.280
1:25 000	1 inch	25 000 in	$\frac{63\,360}{1} = \frac{25\,000}{x}$	0.3945707	2	$\frac{0.3945707}{1} = \frac{2}{y}$	5.068
1:31 250	1 inch	31 250 in	$\frac{63\,360}{1} = \frac{31\,250}{x}$	0.4932133	2	$\frac{0.4932133}{1} = \frac{2}{y}$	4.055
1:31 680	1 inch	31 680 in	$\frac{63\,360}{1} = \frac{31\,680}{x}$	0.500	2	$\frac{0.500}{1} = \frac{2}{y}$	4.000
1:48 000	1 inch	48 000 in	$\frac{63\,360}{1} = \frac{48\,000}{x}$	0.7575757	5	$\frac{0.7575757}{1} = \frac{5}{y}$	6.600
1:50 000	1 inch	50 000 in	$\frac{63\,360}{1} = \frac{50\,000}{x}$	0.7891414	5	$\frac{0.7891414}{1} = \frac{5}{y}$	6.336
1:62 500	1 inch	62 500 in	$\frac{63\,360}{1} = \frac{62\,500}{x}$	0.9864267	6	$\frac{0.9864267}{1} = \frac{6}{y}$	6.082
1:63 360	1 inch	63 360 in	$\frac{63\,360}{1} = \frac{63\,360}{x}$	1.000	6	$\frac{1.000}{1} = \frac{6}{y}$	6.000
1:72 000	1 inch	72 000 in	$\frac{63\,360}{1} = \frac{72\,000}{x}$	1.1363636	6	$\frac{1.1363636}{1} = \frac{6}{y}$	5.280
1:75 000	1 inch	75 000 in	$\frac{63\,360}{1} = \frac{75\,000}{x}$	1.1837121	6	$\frac{1.1837121}{1} = \frac{6}{y}$	5.068
1:96 000	1 inch	96 000 in	$\frac{63\,360}{1} = \frac{96\,000}{x}$	1.5151515	7	$\frac{1.5151515}{1} = \frac{7}{y}$	4.620
1:100 000	1 inch	100 000 in	$\frac{63\,360}{1} = \frac{100\,000}{x}$	1.5782828	9	$\frac{1.5782828}{1} = \frac{9}{y}$	5.702
1:125 000	1 inch	125 000 in	$\frac{63\,360}{1} = \frac{125\,000}{x}$	1.9728535	12	$\frac{1.9728535}{1} = \frac{12}{y}$	6.082
1:150 000	1 inch	150 000 in	$\frac{63\,360}{1} = \frac{150\,000}{x}$	2.3674242	12	$\frac{2.3674242}{1} = \frac{12}{y}$	5.068

To find miles per inch on 1: 12 000 map . . .

63,360 inches = 1 mile
Show in ratio as ...

$$\frac{63\,360}{1} \quad \frac{\textit{inches}}{\textit{miles}}$$

Let SCALE (12 000) be in inches
Fractional scale says 1 inch represents 12,000 in
Let **x** be miles that 1 inch represents on map
Show in ratio as ...

$$\frac{12\,000}{x} \quad \frac{\textit{inches}}{\textit{miles}}$$

Solution . . .

$$\frac{63\,360}{1} = \frac{12\,000}{x}$$

$$63\,360 \cdot x = 12\,000 \cdot 1$$

$$\frac{63\,360\,x}{63\,360} = \frac{12\,000}{63\,360}$$

$$x = \frac{12\,000\ (\text{SCALE})}{63\,360}$$

$$x = 0.1893939$$